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Effective extraction of harmine by menthol/anise alcohol-based natural deep eutectic solvents

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ABSTRACT

Natural deep eutectic solvents composed of two or more natural products are now regarded as green and sustainable alternatives to the conventional solvents and ionic liquids because of the significant advantages of low cost, easy preparation and hypotoxicity. In this work, three kinds of new hydrophobic natural deep eutectic solvents based on natural perfumes, DL-menthol and anise alcohol were prepared and used to extract harmine from aqueous phase. Differences of raw material molar ratios in the preparation lead to the differences in hydrogen bonding of the result products. Hydrogen bonding is the main driving force underlying the extraction, which can be demonstrated by the thermodynamic analysis. The natural deep eutectic solvent with the molar ratio of DL-menthol to anise alcohol at 1 : 1 exhibits high extraction ability for harmine and the optimal extraction conditions are 5.0 min of extraction time, pH 8.0 and 50 : 1 of phase volume ratio (volume of water phase : volume of extractant phase), respectively. Furthermore, the suggested natural deep eutectic solvent can be easily recycled by adjusting pH of the aqueous phase and can be reused at least five times via back extraction. These results indicate that the perfume-based natural deep eutectic solvents are ideal extraction solvents for effectively extracting bioactive compounds.

1. Introduction

Harmine, a β-carboline alkaloid, is widely distributed in plants and the preclinical studies indicate that harmine may exert neuroprotective, cognitive-enhancing and anti-inflammatory effects [1–3]. Harmine is produced predominantly by extraction from the seeds of *Penganum harmala* L. [4,5]. The extraction procedures usually involve the acidification of harmine, followed by the removal of impurities with petroleum ether, and then the basification of the extract, finally the extraction of harmine from the extract with organic solvents (such as chloroform and ethyl acetate) [5–7]. Although higher extraction efficiency of harmine (0.476%, content of harmine in plant sample (weight of harmine/weight of sample)) [7] can be obtained with the use of conventional solvents as extractants, such conventional solvents (e.g., chloroform) are usually toxic [8,9]. Therefore, the development of environmentally benign solvents is of great importance.

In this context, ionic liquids (ILs) entirely composed of ions have attracted much attention as environmentally benign solvents, particularly in the field of separation due to their excellent properties such as negligible vapor pressure [10]. Compared with conventional solvents, ILs exhibit higher extraction ability for alkaloids [11,12]. Despite of the imidazolium-based ILs owing high extraction ability for alkaloids, they have some drawbacks, such as moderate toxicity [13] and complicated synthetic routes (Menshutkin reaction followed by metathesis reaction) [12,14]. Recently, a new type of solvent, deep eutectic solvents (DESs) which are eutectic mixtures of two or more components have been suggested as eco-friendly alternatives to organic solvents and ILs. DESs can be applied as media to the organic extraction, organic and enzymatic reactions [15–18]. Even so, the synthetic components of DESs are toxic and not suitable for applications in food and pharmacology industries. The DESs composed of natural products (NADESs) such as plant constituents (sugars, phenols and organic acids, etc.) have particularly drawn much attention and been widely used to extract bioactive substance due to their better biocompatibility and high solubility for natural products [19–21]. Fernández's team reported a review presenting an overview of knowledge regarding of NADESs-based extraction procedures, focused on reported literature spanning from 2011 up to 2018. Moreover, they tried to find out whether the use of NADESs as extraction media is enough for greening an analytical methodology and to present strategies to make them even greener [20]. Generally, NADESs are formed by a hydrogen-bond acceptor (HBA, for example a quaternary ammonium halide salt) and a hydrogen-bond donor (HBD,

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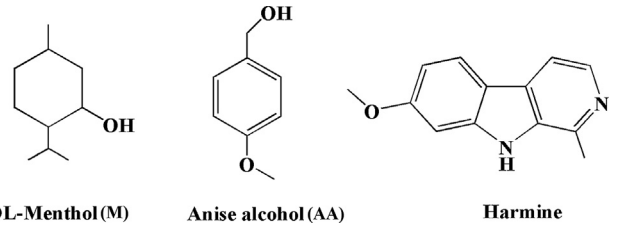


Fig. 1. Chemical structures of DL-menthol (M), anise alcohol (AA) and harmine.

such as amines and carboxylic acids), resulting in hydrogen bonding which is beneficial to the extraction of natural products [15–18]. From the above, NADESs are more promising solvents because they are much cheaper, lower toxicity and easier preparation (100% atom economy) compared with ILs.

Inspired by these excellent studies, in this work, the new hydrophobic NADESs composed of natural perfumes, DL-menthol (M) and anise alcohol (AA) (Fig. 1) were designed and their extraction ability for harmine was investigated. The reason why M and AA are selected as starting materials lies in the fact that they have low water-solubility, low toxicity, biocompatibility and relatively low price (M, about €60 kg^{-1}; AA, \$83 kg^{-1}) [22,23]. Furthermore, both M and AA contain hydroxyl group and the hydrogen bonding between the hydroxyl groups of M and AA implies that the combination of the two compounds may produce new NADESs.

2. Experimental

2.1. Materials

DL-Menthol (M, 98%) and harmine (98%) were purchased from Tokyo Chemical Industry Co., Ltd. (Tokyo, Japan). Anise alcohol (AA, 98%) and methyl *tert*-butyl ether (MTBE, 99%) were obtained from Macklin Biochemical Co., Ltd. (Shanghai, China). Reichardt's dye (RD, 90%) and 4-nitroaniline (NA, ≥99%) were purchased from Sigma-Aldrich Co., (St. Louis, MO, USA); *N*,*N*-diethyl-4-nitroaniline (DENA, 97%) was obtained from Fluorochem Ltd., (Hadfield, UK). All other chemicals used in this work were of analytical grade unless stated. Ultrapure water (18.2 MΩ cm) produced by an Aquapro purification system (Aquapro International Co., Ltd., Dover, USA) was used throughout the experiments.

The seeds of *Penganum harmala* L. was purchased from local drug store and crushed by a disintegrator (model: 08A1, Xulang Co., Guangzhou, China) to minus 100 mesh before extraction.

2.2. Preparation of NADESs

DL-Menthol (M) and anise alcohol (AA) were mixed at a molar ratio of 1 : 1 at 60 °C until a clear and homogeneous liquid formed, which is noted as MAA-I. The NADESs, MAA-II (M : AA = 2 : 1) and MAA-III (M : AA = 1 : 2) were prepared by a similar way.

2.3. Determination of the mutual solubilities of NADESs and water

The solubilities of NADESs in water were measured via the cloud-point method reported in literature [24,25]: one hundred milliliters of water were transferred into a vial and then a small amount of a given NADES (0.9 mg to 8.0 mg) was added. The resulting mixture was stirred with a constant-temperature heating magnetic stirrer (model DF-101S, Yuhua Instrument Co., Ltd., Gongyi, China) at 25 °C. Only if the small amount of a NADES was dissolved after 2 h, the experiment was continued by adding again a weighed amount of a given NADES. When a small amount of fine drops of a given NADES in water was found and lasted for more than 24 h, it was considered that full saturation was achieved.

The solubilities of water in NADESs were determined as follows [26]: equal amounts of a given NADES and water were mixed under stirring for 30 min at 25 °C and then left for 24 h to allow complete equilibration and separation of the two phases. The NADES phase was withdrawn and its water content was measured by a Karl Fischer moisture meter (model KF-1B, Benchuang Analytical Instrument Co., Ltd., Zibo, China).

2.4. Measurements of viscosity, density and melting point

The viscosities and densities of NADESs at 25 °C were measured by a rotary viscometer (model NDJ-8S, Changji Geological Instrument Co., Ltd., Shanghai, China) and a digital density meter (model YT-120G, Yienda Technology Co., Ltd., Xiamen, China), respectively.

The melting points of NADESs, M and AA were determined by differential scanning calorimetry (DSC). The DSC experiments were carried out using the DSC Q2000 instrument (TA Instruments, New Castle, DE, USA) under an argon dry atmosphere with a cooling rate of 5 °C min^{-1} and a heating rate of 1 °C min^{-1}. The obtained melting points (T_m) were used to draw the solid–liquid phase diagram.

2.5. Measurements of proton nuclear magnetic resonance (^{1}H NMR) and fourier transform infrared (FT-IR) spectra

The ^{1}H NMR and FT-IR experiments were conducted using a ^{1}H NMR spectrometer (model Ascend 400, 400 MHz, Bruker Biospin, Rheinstetten, Germany) and a V70 FT-IR spectrophotometer (Bruker Optic GmbH, Ettlingen, Germany), respectively.

2.6. Analysis of harmine by high performance liquid chromatography (HPLC)

The HPLC analysis was performed on an Agilent 1200 HPLC system equipped with an autosampler and a variable-wavelength ultraviolet–visible detector (Agilent Technologies, Santa Clara, CA, USA). Separation was conducted on an Amethyst C18-H column (4.6 mm × 150 mm, 5 μm, Sepax Technologies Inc., Newark, DE, USA) and the column temperature was 30 °C. The mobile phase consisted of 15% (*V*/*V*) acetonitrile and 75% water containing 0.1% (*V*/*V*) acetic acid with a flow-rate of 0.8 mL min^{-1}. The detection wavelength was 318 nm and the injection volume was 10 μL.

2.7. Extraction procedure

Typically, 0.5 mL of a given NADES was mixed with 25 mL of harmine solution (5 mg L^{-1}) at 25 °C for 5 min and phase separation was achieved by centrifugation for 5 min. The harmine concentration in aqueous phase was analyzed by HPLC. The concentration of harmine in NADES phase was calculated by material balance. The extraction efficiency (E) is calculated by the following equation:

$$E = (1 - \frac{Cw}{C_w^i}) \times 100\% \quad (1)$$

where C_w^i and C_w are the concentrations of harmine in water phase before and after extraction, respectively.

The distribution ratio (D) is defined as:

$$D = \frac{Co}{Cw} \quad (2)$$

where C_o and C_w are the harmine concentrations in the NADES and water phases, respectively. To accurately determine E and D values, the NADES phase and aqueous phase were mutually saturated before they were mixed with each other to reduce the changes in phase volume.

2.8. Measurements of Kamlet–Taft polarity parameters

The Kamlet–Taft polarity parameters, α (hydrogen bond donating ability), β (hydrogen bond accepting ability), polarity ($E_T(30)$) and π^* (dipolarity/polarisability) of the NADESs were measured using the three dyes, Reichardt's dye (RD), N,N-diethyl-4-nitroaniline (DENA) and 4-nitroaniline (NA) as probes via measuring their ultraviolet–visible (UV–Vis) maximum absorption wavelengths (λ_{max}). The NA concentrations in the NADESs were 3.3×10^{-5} mol L^{-1} and the DENA concentrations in the NADESs were in the range of 1.7×10^{-5} to 2.1×10^{-5} mol L^{-1}. The concentrations of RD in NADESs ranged from 3.3×10^{-4} to 4.0×10^{-4} mol L^{-1}. The UV–Vis spectra of the three dyes in NADESs were measured by a UV–Vis spectrophotometer (model TU-1810, Purkinje General Instrument Co., Beijing, China). The Kamlet–Taft polarity parameters were calculated by the following equations [27]:

$$\nu_{max}(cm^{-1}) = 10^4/\lambda_{max}(nm) \quad (3)$$

$$E_T(30) = 28592/\lambda_{max}(RD) \quad (4)$$

$$\pi* = 0.314(27.52 - \nu_{max}(DENA)) \quad (5)$$

$$\alpha = 0.0649E_T(30) - 2.03 - (0.72\pi*) \quad (6)$$

$$\beta = (1.035\nu_{max}(DENA) + 2.64 - \nu_{max}(NA))/2.8 \quad (7)$$

where ν_{max}, λ_{max}, $E_T(30)$, π^*, α, β are light frequency corresponding to maximum absorption wavelength, maximum absorption wavelength, solvent polarity, solvent dipolarity/polarisability, solvent hydrogen bond donating ability and solvent hydrogen bond accepting ability, respectively.

2.9. Isolation of harmine from Penganum harmala L.

The isolation of harmine from *Penganum harmala* L. was conducted via the reported method with minor modification [28]: 2.0 g of *Penganum harmala* L. powder was macerated with 50 mL of HCl solution (pH 2.0) for 12 h and then the pH of the aqueous phase was adjusted to 8.0 with ammonia. After filtration, the extract containing harmine was directly extracted by MAA-I at a phase volume ratio of 50 : 1 (V_{water} : V_{MAA-I}) for 5 min. Then the upper phase (NADES phase) was withdrawn and washed by equal volume of HCl solution (pH 2.0). The pH of the washing solution was adjusted to pH 10 with ammonia and the yellow solid containing harmine was obtained. This raw product was washed several times with water to remove the color impurities. The resultant white solid was dissolved with ethanol and further purified with column chromatography (stationary phase: silica gel (200–300 mesh); mobile phase: ethanol and ethyl acetate (1 : 8, *V/V*)).

All the above experiments were conducted in triplicate and all data presented in this work are expressed as mean ± standard deviation (SD).

3. Results and discussion

3.1. Characterization of NADESs

Generally, NADESs are composed of HBD and HBA, which can be associated with each other through strong hydrogen bonding [15,16]. In order to identify HBD and HBA components of the synthesized NADESs, the ^{1}H NMR experiments were performed and the results are illustrated in Fig. 2 (details about ^{1}H NMR spectra including chemical shifts and integration values are shown in Fig. S1 to Fig. S5 in the Supporting Information (SI)). It can be clearly seen from Fig. 2 that after mixing, the ^{1}H NMR signals of the hydroxyl protons of both M and AA components are completely overlapped, suggesting that the hydroxyl protons are in exchange between the two components. Meanwhile, after mixing with AA, the chemical shift value of the hydroxyl

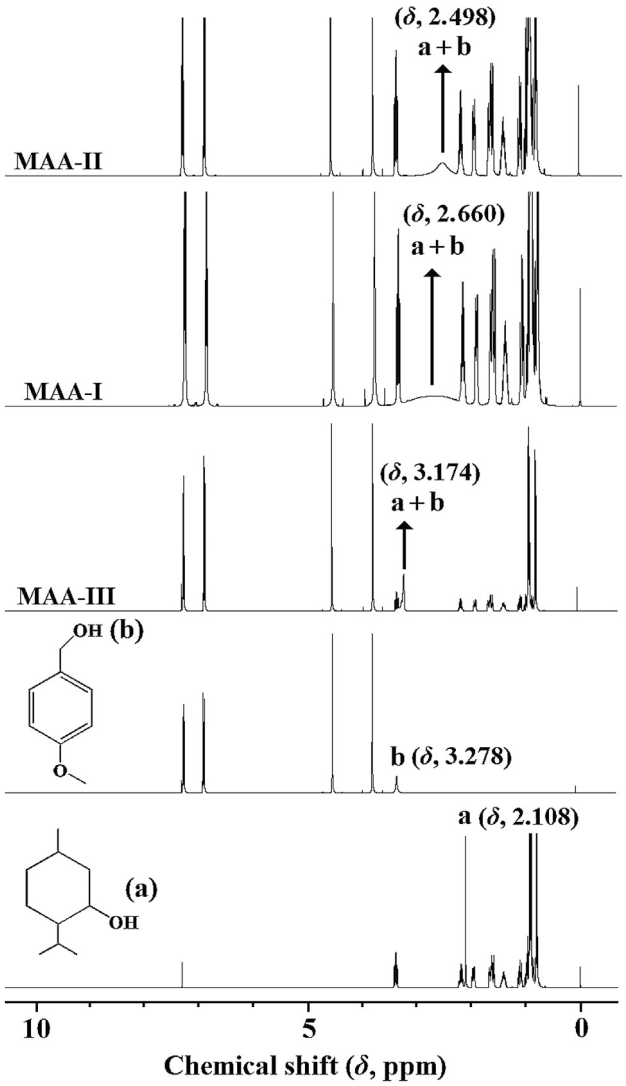


Fig. 2. The ^{1}H NMR spectra of M, AA, MAA-I, MAA-II and MAA-III (400 MHz, solvent, $CDCl_3$).

proton of M increases, which is the typical characteristic of HBD (the formation of hydrogen bond will decrease the electron density around the hydrogen nucleus of HBD, leading to deshielding effect) [29,30]. As far as AA is concerned, after mixing with M, the chemical shift value of its hydroxyl proton decreases. This is the typical feature of HBA (the formation of hydrogen bond leads to the increase of the electron density of the proton acceptor oxygen (i.e., hydroxyl oxygen of AA), resulting in shielding effect) [29,30]. These observations suggest that there exists hydrogen bonding between M and AA and the starting materials M and AA act as HBD and HBA, respectively.

Additionally, to analyze the hydrophobic nature of the synthesized NADESs, ^{1}H NMR technique was used to monitor the mutual solubilities of NADESs and water. As shown in Fig. S3 to Fig. S6 (see SI), after contacting with water, there only weak water signal can be observed in the NADES-rich phase. Besides water signal, there no obvious changes in other ^{1}H NMR signals are observed, suggesting the good stability of NADESs upon contact with water. The ^{1}H NMR signals of NADESs in water phase (D_2O) is also weak. These results indicate that the mutual solubilities of NADESs and water are very limited. Furthermore, the solubilities of NADESs in water and the solubilities of water in NADESs were quantitatively determined by cloud-point method [24,25] and Karl Fischer method [26], respectively. The results listed in Table 1 show that the solubilities of water in NADESs (saturated water contents

Table 1
The Kamlet–Taft polarity parameters, viscosities, densities, solubilities and water contents of the synthesized NADESs (mean ± standard deviation, n = 3).

NADES	$E_T(30)$	π^*	α	β	$\alpha\beta$	Viscosity (mPa·s)	Density (g mL^{-1})	Solubility in water (mg (100 mL)$^{-1}$)	Saturated water content (wt.%)
MAA-I	48.2 ± 0.037	0.76 ± 0.0069	0.55 ± 0.0055	0.61 ± 0.016	0.34 ± 0.011	64.1 ± 0.15	0.991 ± 0.00058	38.3 ± 1.3	0.22 ± 0.013
MAA-II	47.1 ± 0.016	0.70 ± 0.0033	0.52 ± 0.0026	0.64 ± 0.010	0.33 ± 0.0061	66.2 ± 0.38	0.953 ± 0.0035	30.9 ± 0.85	0.13 ± 0.0050
MAA-III	48.7 ± 0.041	0.88 ± 0.0025	0.50 ± 0.0033	0.57 ± 0.0085	0.29 ± 0.0050	66.9 ± 0.26	1.017 ± 0.0012	52.8 ± 0.71	0.29 ± 0.021

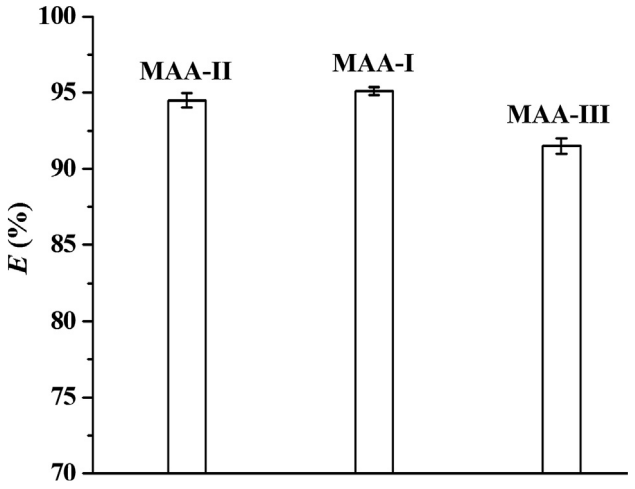


Fig. 3. Extraction efficiency (E) of the NADESs composed of M and AA in different molar ratios. Extraction conditions: V_W : V_{NADES} = 50 : 1; $C_{harmine}$ = 5 mg L^{-1}; extraction time, 5 min; extraction temperature, 25 °C; pH of aqueous phase, 8.0. Data are listed in Table S2 in the SI.

of NADESs) range from (0.13 ± 0.0050)% to (0.29 ± 0.021)% and the solubilities of NADESs in water are in the range of (30.9 ± 0.85) mg (100 mL)$^{-1}$ to (52.8 ± 0.71) mg (100 mL)$^{-1}$, suggesting very limited mutual solubility between water and NADESs.

Finally, the physical properties of the synthesized NADESs, including their viscosities, densities, and Kamlet–Taft polarity parameters are listed in Table 1. The phase diagram is shown in Fig. S7 in the SI (melting points of the mixtures of M and AA are listed in Table S1 in the SI). As can be seen from Fig. S7 and Table S1, the melting points of the mixtures of M and AA are lower than those of the starting materials (M and AA), suggesting the formation of NADESs.

3.2. Extraction ability of MAA-I MAA-II and MAA-III

In this work, NADESs with different M/AA ratios were used for the extraction of harmine from the aqueous phase. The results shown in Fig. 3 indicate that MAA-I (M : AA = 1 : 1) and MAA-II (M : AA = 2 : 1) have roughly the same extraction efficiency for harmine and the extraction efficiency of MAA-III (M : AA = 1 : 2) is slightly lower than the above two NADESs, which may be ascribed to hydrogen bonding. Recently, much work reported that hydrogen bonding played an important role in the extraction of alkaloids and organic acids [11,31–33]. To explore the driving force underlying the extraction process, thermodynamic analysis was conducted because the values of thermodynamic parameters, enthalpy change (ΔH) and entropy change (ΔS) are the indicators of the driving force in the extraction process: $\Delta H < 0$ and $\Delta S < 0$ mean hydrogen bonding; $\Delta H > 0$ and $\Delta S > 0$ suggest hydrophobic interaction; $\Delta H < 0$ and $\Delta S > 0$ indicate electrostatic force [34,35]. The values of ΔH and ΔS can be obtained using the van't Hoff equation [34,35]:

$$\ln D = -\frac{\Delta H}{RT} + \frac{\Delta S}{R} \tag{8}$$

where R and T are the gas constant and thermodynamic temperature, respectively.

The van't Hoff curves of the NADESs are shown in Fig. 4 and the resulting ΔH and ΔS values are shown in Table 2. It is obvious that the ΔH and ΔS values for the extraction of harmine by the three NADESs are both negative, suggesting that hydrogen bonding is the main driving force underlying the extraction process [34,35]. To further confirm this deduction, FT-IR experiments were conducted and the FT-IR spectra of MAA-I/harmine system, as an example, are shown in Fig. 5. The other FT-IR spectra are illustrated in Fig. S8 and Fig. S9, respectively (see SI).

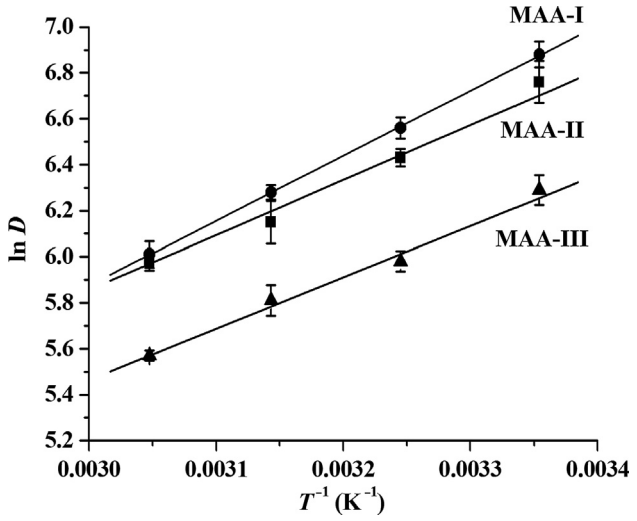


Fig. 4. The van't Hoff curves of the extraction of harmine by the NADESs. Extraction conditions: V_W : V_{NADES} = 50 : 1; $C_{harmine}$ = 5 mg L^{-1}; extraction time, 5 min; pH of aqueous phase, 8.0. Data are listed in Table S3 in the SI.

Table 2
The ΔH and ΔS values of the extraction of harmine by NADESs (mean ± standard deviation, n = 3).

NADES	ΔH (kJ mol^{-1})	ΔS (J mol^{-1} K^{-1})
MAA-I	−(23.5 ± 3.4)	−(21.7 ± 11)
MAA-II	−(21.4 ± 4.2)	−(15.7 ± 13)
MAA-III	−(19.0 ± 3.1)	−(11.5 ± 9.8)

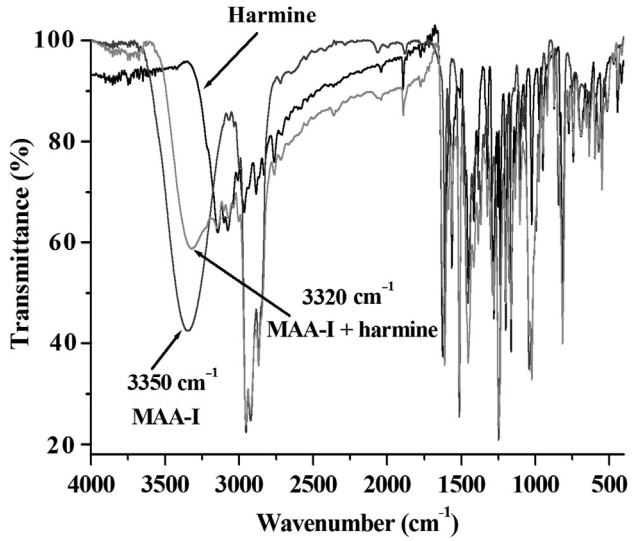


Fig. 5. The FT-IR spectra of harmine, MAA-I and their mixture.

As can be seen from Fig. 5, the NADES, MAA-I has a representative band corresponding to the hydroxyl group, at 3350 cm^{-1} after contacting with harmine, the hydroxyl band of MAA-I shifts to lower wavenumber value (3320 cm^{-1}), indicating the formation of a new hydrogen bond between MAA-I and harmine (the formation of hydrogen bond decreases the energy of the O—H bond, causing it to vibrate at lower frequency) [36,37]. Similar phenomena are also observed for the MAA-II/harmine and MAA-III/harmine systems (Fig. S8 to Fig. S9 in the SI).

Since the extraction of harmine is main driven by hydrogen bonding, the Kamlet–Taft polarity parameters of the three NADESs were measured. The product of α with β ($\alpha\beta$) has been recommended by recent work as an indicator of the strength of hydrogen bonding [38–40]. This suggestion is adopted in the present work. As can be seen from Table 1, the hydrogen bonding strength ($\alpha\beta$ value) of MAA-I and MAA-II are similar and higher than MAA-III, which explains why MAA-III has lower extraction ability for harmine. Based on the above discussion, MAA-I is selected as the extraction solvent and used in the following studies due to the lower cost of anise alcohol.

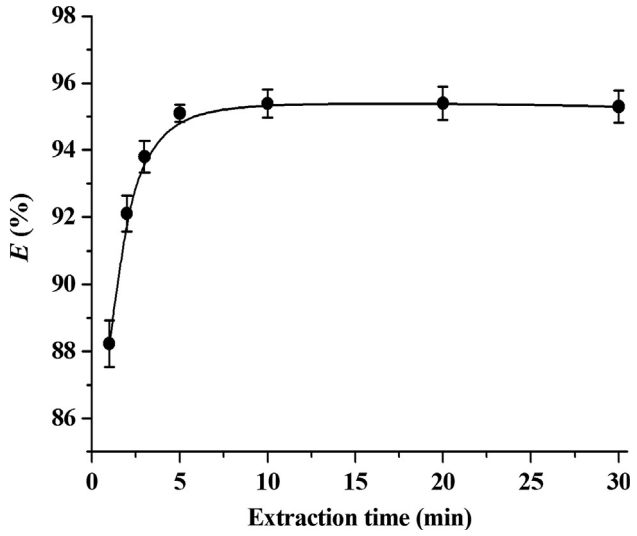


Fig. 6. Effect of the extraction time on the extraction efficiency. Extraction conditions: V_W : V_{MAA-I} = 50 : 1; $C_{harmine}$ = 5 mg L^{-1}; extraction temperature, 25 °C; pH of aqueous phase, 8.0. Data are listed in Table S4 in the SI.

3.3. Optimization of the extraction conditions

In this work, the extraction conditions including extraction time, phase volume ratio, extraction temperature and pH of the aqueous phase were optimized. As shown in Fig. 6, the extraction efficiency of harmine increases from 1 min to 5 min and then keeps constant with prolonging the extraction time, which means that the extraction of harmine can reach equilibrium quickly. Based on these results, 5 min is regarded as the optimal extraction time.

The influence of pH on the extraction of harmine is shown in Fig. 7. It is clear that the extraction efficiency increases with increasing the pH

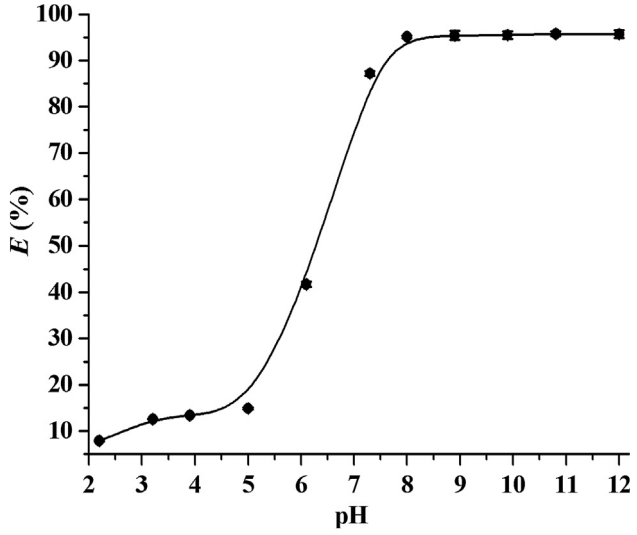


Fig. 7. The influence of aqueous pH on the extraction of harmine. Extraction conditions: V_W : V_{MAA-I} = 50 : 1; $C_{harmine}$ = 5 mg L^{-1}; extraction temperature, 25 °C; extraction time, 5 min. Data are listed in Table S5 in the SI.

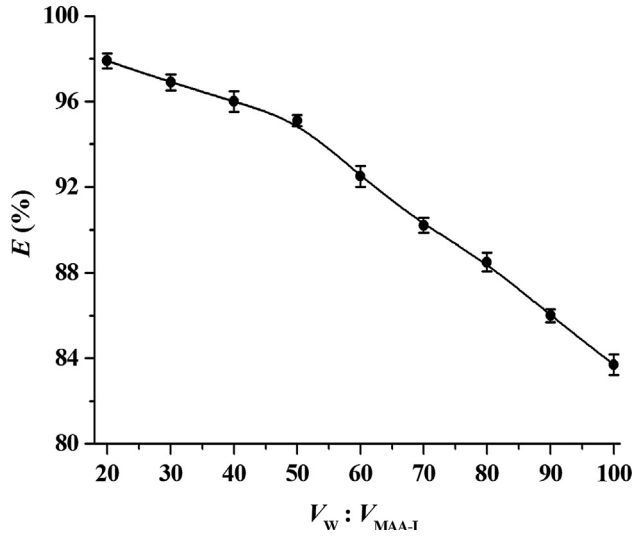


Fig. 8. The influence of phase volume ratio on the extraction of harmine. Extraction conditions: $C_{harmine}$ = 5 mg L^{-1}; extraction temperature, 25 °C; extraction time, 5 min; pH of aqueous phase, 8.0. Data are listed in Table S6 in the SI.

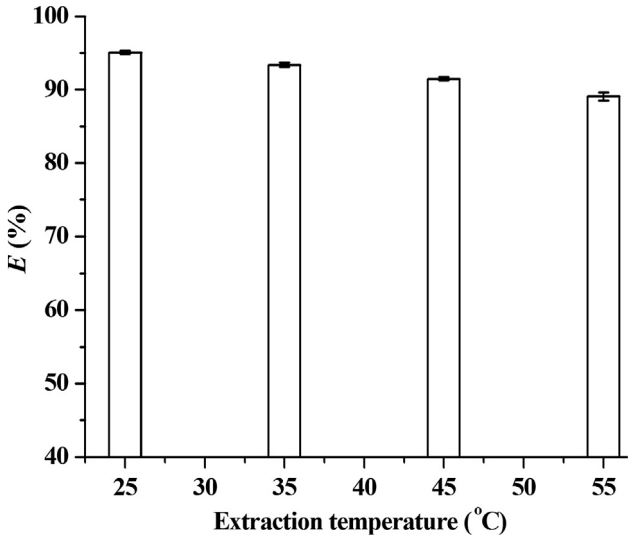


Fig. 9. The influence of temperature on the extraction of harmine. Extraction conditions: $V_W : V_{MAA-I}$ = 50 : 1; $C_{harmine}$ = 5 mg L^{-1}; pH of aqueous phase, 8.0; extraction time, 5 min. Data are listed in Table S7 in the SI.

from 2.0 to 8.0 and then maintains constant. It is known that the dissociation constant (pK_a) of harmine is 7.45 [41]. This means that when pH > 7.45, harmine exists in the neutral form. When pH < 7.45, the nitrogen atom in harmine will be protonated and the cationic form is predominant. The protonation of nitrogen atom reduces the hydrogen bonding between harmine and NADES and decreases the extraction efficiency accordingly [31,42]. Based on these results, 8.0 is regarded as the optimal pH value of the aqueous phase.

The volume ratio of water to NADES phases ($V_w : V_{MAA-I}$) is another important parameters in the extraction. High phase volume ratio means the reduction of the consumption of NADESs, which can reduce the operation cost. As shown in Fig. 8, the extraction efficiency of harmine decreases with the increase of phase volume ratio, which is in good agreement with the relationship of extraction efficiency (E) and phase volume ratio ($V_w : V_{MAA-I}$).

$$E = \frac{D}{D + \frac{V_w}{V_{MAA-I}}} \times 100\% \quad (9)$$

Considering the fact that the phase volume ratio, $V_w : V_{MAA-I}$ = 50 : 1, can provide higher extraction efficiency (> 95%), it is selected as the optimal phase volume ratio and used through the experiments.

The effect of temperature on the extraction efficiency of harmine is shown in Fig. 9. As can be seen, the increase of temperature only leads to a slight decrease of the extraction efficiency, meaning that the extraction can be conducted over a wide temperature range. In view of the fact that extraction at 25 °C has the highest extraction efficiency, this temperature is regarded as the optimal extraction temperature.

3.4. Compared with the reported methods

The conventional solvents chloroform [6,7], ethyl acetate [5] and the mixture of methyl *tert*-butyl ether (MTBE) and tetrahydrofuran (THF, $V : V$ = 1 : 1) containing 10^{-2} mol L^{-1} of triethylamine (TEA) (MTBE/THF/TEA) [4] have been used to extract harmine from aqueous phase. As mentioned above, ILs have been widely used to extract alkaloids [11,12], but there is no reported literature on the extraction of harmine by ILs. Our previous work suggests that the ILs, 1-butyl-3-decylimidazolium trifluoromethanesulfonate ($[C_4C_{10}im]CF_3SO_3$) and 1-methoxyethyl-3-buytlimidazolium bis(trifluoromethanesulphonyl) imide ($[C_4C_1OC_2im]NTf_2$) exhibit high extraction ability for the alkaloids quinine [31], gramine [31] and hordenine [11]. Therefore, $[C_4C_{10}im]CF_3SO_3$ and $[C_4C_1OC_2im]NTf_2$ were selected as the representatives of ILs to explore whether the ILs have comparable extraction performance with the NADES, MAA-I. Considering the fact that water and the extractants (ILs, MAA-I, chloroform, ethyl acetate, and MTBE/THF/TEA) have a certain mutual solubility, water phase and extractant phase were mutually saturated with each other before extraction to avoid the changes in phase volume. The results shown in Fig. 10 indicates that MAA-I has the strongest extraction ability ($V_{water} : V_{extractant}$ = 50 : 1). Furthermore, chloroform is carcinogenic [8] and the dialkylimidazolium-based ILs are expensive due to their complex synthetic procedures [12,14]. As a contrast, MAA-I can be easily prepared with 100% atom economy. Furthermore, the toxicity of NADESs is significantly lower than the dialkylimidazolium-based ILs [21]. These results suggest that NADESs fully comply with the demands of green chemistry principles and are more suitable for industrial and laboratory applications.

3.5. The reuse of MAA-I and the purification of harmine

To recycle the MAA-I, after extraction, back extraction was

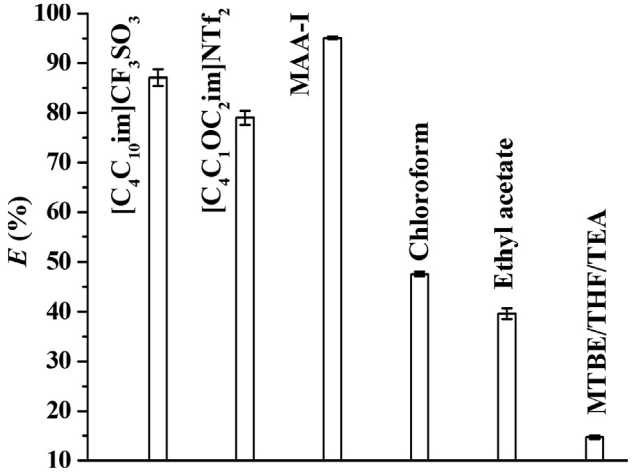


Fig. 10. The extraction ability of different systems for harmine. Extraction conditions: $C_{harmine}$ = 5 mg L^{-1}; extraction temperature, 25 °C; extraction time, 5 min; pH of aqueous phase, 8.0; $V_W : V_{extractant}$ = 50 : 1. Data are listed in Table S8 in the SI.

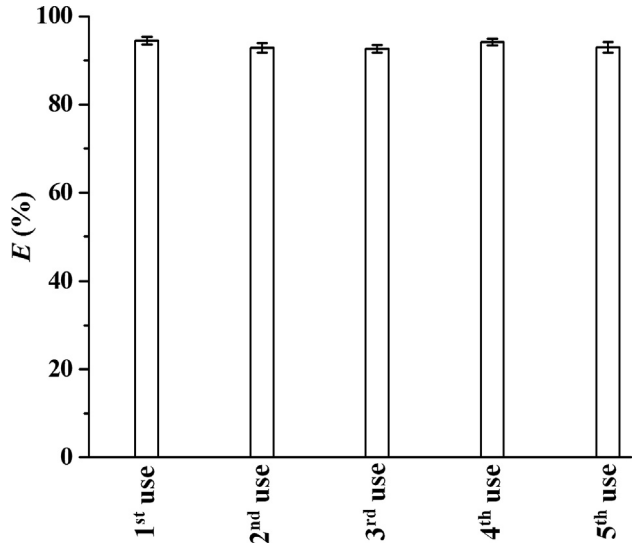


Fig. 11. Reusability of MAA-I for the extraction of harmine from the extract of seeds of *Penganum harmala* L.. Experimental conditions: extraction temperature, 25 °C; extraction time, 5 min; pH of aqueous phase, 8.0; $V_W : V_{MAA-I} = 50 : 1$. Data are listed in Table S9 in the SI.

conducted via washing the MAA-I phase with equal volume of HCl solution (pH 2.0). The recovered NADES phase can be reused at least five times to extract harmine from the extract of seeds of *Penganum harmala* L. without losing its extraction ability (Fig. 11). It should be noted that the solubilities of the suggested NADES, MAA-I, in the extract of seeds of *Penganum harmala* L. and HCl solution (pH 2.0) are (33.7 ± 1.4) mg (100 mL) $^{-1}$ and (36.8 ± 1.7) mg (100 mL) $^{-1}$, respectively, indicating that the loss of MAA-I from dissolution is very limited during the reuse and extraction processes. Additionally, after back extraction, the purity of the raw product is (81.4 ± 1.6)% (harmine content). Therefore, the raw product was further purified by column chromatography and a purity of harmine as high as (97.5 ± 0.93)% was obtained (determined by the aforementioned HPLC method).

4. Conclusions

In the present study, new green natural deep eutectic solvents (NADESs) based on the mixtures of DL-menthol (M) and anise alcohol (AA) in different proportions were prepared via mixing method and their extraction ability for harmine was systematically investigated. Subsequent optimization for the extraction of harmine indicated that the extraction reached equilibrium quickly (5 min), could be operated at a high phase volume ratio ($V_{water} : V_{NADES} = 50 : 1$) and was mainly driven by hydrogen bonding. After extraction, MAA-I could be reused at least five times via back extraction with HCl solution. The extraction ability of MAA-I for harmine was higher than the traditional organic solvents (ethyl acetate, MTBE/TEA and chloroform) and ILs ($[C_4C_{10}im]CF_3SO_3$ and $[C_4C_1OC_2im]NTf_2$). Besides, MAA-I possessed the advantages of ease to prepare, low cost and more greener. The above findings are expected to extend the applications of NADESs as promising green and effective solvents for the extraction of bioactive compounds from plants.

CRediT authorship contribution statement

Yunchang Fan: Conceptualization, Methodology, Writing - original draft. **Hongwei Wu:** Conceptualization, Investigation. **Dongxu Cai:** Data curation, Investigation. **Tuojie Yang:** Investigation. **Lei Yang:** Writing - review & editing, Conceptualization, Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

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